Advanced Programming Practices
MTech 2026
Lab - 11

General Instructions:

- Write one .c file per problem in the assignment, renamed to RollNo_APP11_Qn.c (e.g. ABC2026001_APP11_Q1.c).
- Every program must compile without warnings using the gcc compiler.
- Free any memory you allocate. Programs that leak memory or access out-of-bounds indices will lose marks even if the visible output looks correct.
- Test each program with at least one normal case, one edge case (e.g. size 0 or 1, all-equal elements), and one larger randomly generated case.
- Submit all the .c files on Google Classroom before the due time, any late submissions will be penalized or discarded.

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1. Write a menu driven program for performing stack operations such a PUSH, POP, finding top of the stack.
2. Design a recursive function in C language that reverses the stack using only stack itself, without using another stack, array, or linked list. Your solution must use recursion to remove the top element and insert it at the bottom.